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Digital-Optical Implementation of Quantum-Inspired and Classical Classifiers with a Joint Transform Correlator

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Abstract

We explore the implementation of quantum-inspired and classical distance-based classifiers using an optical architecture known as the Joint Transform Correlator (JTC). These classifiers, including the classical and quantum nearest-mean classifier (CNM-C, QNM-C) and radial basis function classifier (RBF-C), rely on distance computations between a query sample and class prototypes. While traditionally executed in software, such operations become computationally intensive in high dimensions or large datasets. We propose a new optical implementation using a reflective, phase-only, spatial light modulator (SLM) and a CMOS camera within a 1-f Fourier setup, leveraging the speed and parallelism of coherent light. We detail the theoretical framework linking optical correlation with similarity measures used in classification, and describe the data encoding strategies that map feature vectors to optical patterns.

I. Introduction

Quantum-inspired algorithms are classical routines that use mathematical structures and physical intuitions from quantum mechanics — such as superposition-style probability amplitudes, interference or quantum state discrimination — yet run entirely on conventional hardware. These algorithms have multiple applications, including machine learning, where they promise faster convergence or reduced dimensional dependence compared with traditional methods.

Some machine-learning algorithms are based on distance. Here, the decision rule depends on the similarity between a query sample and a set of prototypes, typically evaluated through dot products or euclidean distance calculation. Although straightforward in software, these operations become a computational bottleneck for high-dimensional data streams or large datasets. There have been optical implementations of classical classification algorithms ([6], [4]). These implementations offer a compelling alternative: coherent light can perform calculations at the speed of propagation, exploiting spatial parallelism while consuming only milliwatts of power.

Among the various optical architectures, the *Joint Transform Correlator* (JTC) ([7], [3]) stands out for its simplicity and adaptability. Unlike classical VanderLugt systems ([9]), the JTC does not require a pre-fabricated filter; instead, the reference and query patterns are placed side by side in the input plane. A single Fourier transform—implemented with a lens—produces an output intensity whose off-axis terms encode the cross-correlation between the two patterns. When feature vectors are encoded as two-dimensional phase distributions, the correlation peak height provides a direct measure of their similarity, which can be mapped to a distance-based decision rule.

In this work we propose optical implementations of the Quantum-Inspired and Classical Nearest Mean Classifier ([2], [8]), and Radial Basis Function Classifier ([6], [4]), that use a single SLM and a 1-f lens system ([5]). We begin by reviewing the theoretical connection between optical correlation and distance-based similarity measures. We then describe our experimental set-up, which integrates a single reflective, phase only, spatial light modulator and a CMOS camera to implement a a new version of the QNM-C (Section **??**). Finally, we benchmark the models with the MNIST dataset and compare their performances with purely electronic implementations, highlighting the trade-offs (Section V).

II. Supervised learning and quantum-inspired classification

i. Supervised learning

Machine learning (ML) provides algorithmic tools that *learn* directly from data rather than relying on hand-crafted rules. In the *supervised* setting, learning is guided by *examples* whose correct answers are known in advance. We review the key ingredients used throughout this work.

Input data. We start from N *raw samples* $\mathbf{u}_i \in \mathbb{R}^p$, $i = 1, \ldots, N$, each accompanied by a *label* $y_i \in \{1, \ldots, K\}$. The labels partition the data into

$$C_k = \{\mathbf{u}_i : y_i = k\}, \qquad N_k = |C_k|, \qquad p_k = \frac{N_k}{N},$$

where N_k and p_k are the size and empirical probability of class k, respectively. It is convenient to organise the entire data set as a matrix $\mathbf{M} = [\mathbf{u}_1 \ \mathbf{u}_2 \ \cdots \ \mathbf{u}_N] \in \mathbb{R}^{p \times N}$ and the labels as a vector $\mathbf{y} = [y_1, \ldots, y_N]^\top$.

Training and test sets. A *training set* $(\mathbf{M}_{\text{train}}, \mathbf{y}_{\text{train}})$ is used to tune the parameters of a classifier, while a disjoint *test set* $(\mathbf{M}_{\text{test}}, \mathbf{y}_{\text{test}})$ gauges its ability to generalise to unseen data. The algorithm is trained on the training set and evaluated on the test set. In this work we will be using the MNIST dataset (Fig.1), with 60000 training samples and 10000 testing samples.

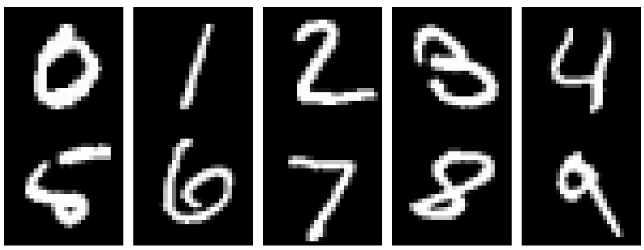


Figure 1: One representative example per class from the MNIST dataset. Each 28×28 grayscale image is flattened to a $p = 784$-dimensional feature vector before the pre-processing pipeline (centering, standardisation, PCA).

Pre-processing. Real-world data rarely arrive in a form that is optimal for learning. We can apply three light transformations that are standard in pattern recognition:

(a) *Centring* removes global offsets, $\mathbf{u}_i \longmapsto \mathbf{u}_i - \bar{\mathbf{u}}$, where $\bar{\mathbf{u}} = N^{-1} \sum_{j=1}^{N} \mathbf{u}_j$.

(b) *Standardisation* rescales each feature so that every dimension has unit variance, preventing attributes with large numerical ranges from dominating the learning process.

(c) *Principal-component analysis* (PCA) projects the centred, standardised vectors onto the $p'(< p)$ directions of greatest variance, $\mathbf{x}_i = \Pi_{\text{PCA}} \mathbf{u}_i \in \mathbb{R}^{p'}$, thereby discarding redundant noise and shrinking the effective dimensionality.

For our experiments, we will only use PCA.

Balanced accuracy. Performance is summarised through the *confusion matrix* $V_{k'k}$, whose entry counts how many test samples belonging to class k are classified as k'. The *balanced accuracy* (BA)

$$\text{BA} = \frac{1}{K} \sum_{k=1}^{K} \frac{V_{kk}}{N_k}$$

computes the mean of the per-class accuracies and is robust to class imbalance. When classes are homogeneous in size, BA reduces to the usual accuracy.

Classical Nearest-Mean Classifier (CNM-C). The nearest-mean classifier (NMC) is a simple, interpretable classifier that assigns a label to a test sample based on the *centroid* of the training samples in each class (Fig. 2). The *centroid* of class k is defined as the mean of the training samples in that class:

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{u}_i, \qquad \text{where } \mathbf{u}_i \in C_k.$$

The CNM classifier assigns a test sample $\mathbf{x}$ to the class j if:

$$d(\mathbf{x}, \boldsymbol{\mu}_j) = \min_{k=1,\ldots,K} d(\mathbf{x}, \boldsymbol{\mu}_k),$$

where the distance $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ is the Euclidean distance. The CNM classifier is simple to implement, as it only requires the computation of the centroids and K distance calculations, one for each centroid.

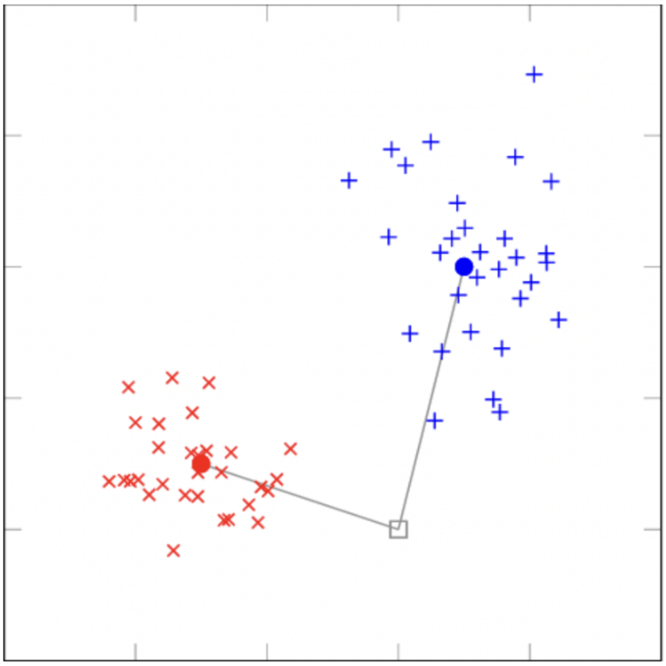


Figure 2: Ilustration of the CNM-C decision rule. The training set is split into two classes and the centroids are computed. The nearest centroid to the test sample is selected as the predicted class.

Radial-Basis-Function Classifier (RBF-C). The RBF model represents a target function as a weighted sum of Gaussian basis functions, each centered at a specific location in the input space. These Gaussian functions produce localized responses that decrease with distance from their centers, allowing the model to approximate smooth and continuous mappings. By tuning the centers, widths, and weights of the basis functions, the RBF network can adapt to complex data distributions and interpolate between training samples with controlled generalization.

We can build a simple RBF network, shown in Fig.3, that contains an input layer of raw features, a hidden layer of M Gaussian units, and a linear output layer that produces one score per class. Every hidden unit returns a similarity value $\varphi_i(\mathbf{x}) = \exp\left[-\|\mathbf{x} - \mathbf{t}_i\|^2 / (2\sigma_i^2)\right]$ to its centre $\mathbf{t}_i$ with width σ_i, so the class score is a weighted sum $f_c(\mathbf{x}) = \sum_{i=1}^{M} w_{c,i}\, \varphi_i(\mathbf{x})$.

Training. Learning splits naturally into two parts:

(a) *Centre and width selection.* Mini-batch k-means clusters the training set; the cluster centroids become the RBF centres $\mathbf{t}_i$. Each width is set to $\sigma_i = k_\sigma \bar{d}_i$, where $\bar{d}_i$ is the average distance to the nearest neighbouring centre, with a floor $\sigma_{\min}$ to avoid underflow.

(b) *Output-weight estimation.* Defining the design matrix $\Phi_{n,i} = \varphi_i(\mathbf{x}_n)$, we solve the ridge-regularised normal equations $\mathbf{W} = (\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top \mathbf{T}$, where $\mathbf{T}$ is the one-hot label matrix. This closed-form step replaces slow gradient-descent procedures and finishes in a single pass.

Inference. For a test vector $\mathbf{x}$, RBF-C computes the hidden activation vector $\boldsymbol{\varphi}(\mathbf{x}) = \left[\varphi_1(\mathbf{x}), \ldots, \varphi_M(\mathbf{x})\right]^\top$, then evaluates $\mathbf{f}(\mathbf{x}) = \mathbf{W}\boldsymbol{\varphi}(\mathbf{x})$ and assigns the class with a winner-takes-all rule: $c^\star = \arg\max_c f_c(\mathbf{x})$. Because the heavy computation is the M distance evaluations, we replace the Euclidean metric with the optical calculated distance, allowing these distances to be calculated faster and more energy-efficiently.

ii. Quantum-inspired classification

Quantum-inspired learning borrows mathematical objects native to quantum theory—*density operators*, *trace distance*, *superposition*—but applies them on ordinary CPUs/GPUs. By embedding classical feature vectors in a Hilbert space, one can exploit the geometry of quantum states to obtain alternative similarity measures that sometimes improve accuracy or robustness compared with purely classical methods [8]. Every quantum-inspired algorithm must start with a function that maps

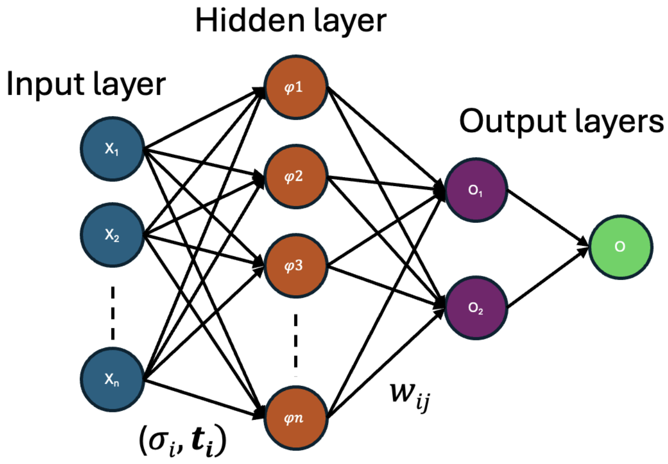


Figure 3: Architecture of the RBF network. The input layer feeds into a hidden layer of radial units, each centered at $\mathbf{t}_i$ with width σ_i. The output layer combines the gaussian responses through learned weights w_{ij}, followed by a winner-takes-all decision stage to determine the predicted class.

each classical data vector x to a density operator ρ_x. The choice of this encoding is very important, as it determines the properties of the resulting quantum-inspired algorithm.

Density–pattern encodings. All quantum-inspired classifiers require a map that sends a real feature vector $\mathbf{x} \in \mathbb{R}^d$ to a *density operator* $\rho_{\mathbf{x}} = \tilde{\mathbf{x}}\,\tilde{\mathbf{x}}^\top$. There are different ways of doing this mapping:

(a) Standard amplitude (SA) encoding.
Normalise the input vector to unit length:

$$\tilde{\mathbf{x}}^{(\mathrm{SA})} = \begin{cases} \dfrac{\mathbf{x}}{\|\mathbf{x}\|}, & \mathbf{x} \neq \mathbf{0}, \\ (1, 0, \dots, 0)^\top, & \mathbf{x} = \mathbf{0}. \end{cases}$$

The encoded state remains in the original d-dimensional Hilbert space.

(b) Inverse stereographic (IS) encoding.
Embed $\mathbf{x}$ into $\mathbb{R}^{d+1}$ through the inverse stereographic projection onto the unit sphere:

$$\tilde{\mathbf{x}}^{(\mathrm{IS})} = \frac{1}{1 + \|\mathbf{x}\|^2}\Big(2x_1, \dots, 2x_d, \|\mathbf{x}\|^2 - 1\Big)^\top.$$

This encoding adds an extra dimension to the input vector, which is then projected onto the unit sphere.

(c) Informative (INF) encoding [8].
Attach the norm as an extra amplitude so that both direction and length of $\mathbf{x}$ affect the representation:

$$\tilde{\mathbf{x}}^{(\mathrm{INF})} = \begin{cases} \dfrac{1}{\sqrt{2}}\Big(\dfrac{\mathbf{x}}{\|\mathbf{x}\|}, 1\Big)^\top, & \mathbf{x} \neq \mathbf{0}, \\ (0, \dots, 0, 1)^\top, & \mathbf{x} = \mathbf{0}. \end{cases}$$

For any of the above encodings, the corresponding *density operator* is obtained by taking the outer product of the encoded vector with itself:

$$\rho_{\mathbf{x}} = \tilde{\mathbf{x}}\,\tilde{\mathbf{x}}^\top.$$

This guarantees $\rho_{\mathbf{x}}$ is positive semidefinite, unit-trace, and idempotent ($\rho_{\mathbf{x}}^2 = \rho_{\mathbf{x}}$), hence a valid pure quantum state. They differ in how much geometric information (direction, norm) is retained, a factor that can significantly affect classification performance.

Quantum centroid. For each class k with training subset $\mathcal{S}_k^{\mathrm{tr}}$, define its *quantum centroid* as the sum of the density operators from the class, divided by the number of elements in the class:

$$\varrho_k = \frac{1}{|\mathcal{S}_k^{\mathrm{tr}}|} \sum_{(\mathbf{x}_i, k) \in \mathcal{S}_k^{\mathrm{tr}}} \rho_{\mathbf{x}_i}.$$

Unlike the classical centroid, ϱ_k is usually a *mixed* state and has no direct counterpart in the original feature space.

Quantum distance measure. Similarity between two density patterns is quantified by the *trace distance*,

$$d_{\mathrm{tr}}(\rho, \sigma) = \tfrac{1}{2}\,\mathrm{Tr}\big|\rho - \sigma\big|,$$

where $|A| = \sqrt{A^\dagger A}$. Note that the trace distance defines a metric; it satisfies the triangle inequality, $d_{tr}(\rho, \sigma) \geq 0$ and $d_{tr}(\rho, \sigma) = d_{tr}(\sigma, \rho)$, and is equal to zero if and only if $\rho = \sigma$.

Having defined the quantum centroid and quantum distance, we can now define the quantum-inspired version of the nearest-mean classifier.

Quantum-inspired nearest-mean classifier (QNM-C). The quantum-inspired nearest-mean classifier (QNM-C) is the exact same as the CNM-C, but uses the quantum centroid and quantum distance instead of the classical centroid and Euclidean distance. The algorithm is as follows:

(a) Encode every training sample $\mathbf{x}_i$ as a density operator $\rho_{\mathbf{x}_i}$ using one of the encodings described in ii.

(b) Compute the quantum centroid for each class k:

$$\varrho_k = \frac{1}{N_k} \sum_{i=1}^{N_k} \rho_{\mathbf{x}_i}.$$

(c) For each test density operator $\rho_{\mathbf{x}}$, assign the class k if

$$d_{\mathrm{tr}}(\rho_{\mathbf{x}}, \varrho_k) = \min_{j=1,\dots,K} d_{\mathrm{tr}}(\rho_{\mathbf{x}}, \varrho_j).$$

Having defined both the classical and quantum-inspired nearest-mean classifiers, we can now describe the joint transform correlator (JTC) and how it can be used to implement these classifiers optically.

III. Joint Transform Correlator

i. Classical Joint Transform Correlator

A *joint transform correlator* (JTC) is an optical architecture that performs the cross– and auto–correlation of two 2-D patterns by means of only *free-space propagation* and *Fourier transform* lenses, thereby exploiting the inherent parallelism and picosecond-scale speed of coherent light. Unlike classical Vander Lugt correlators, the JTC dispenses with a pre-fabricated matched filter: both the *reference* pattern $s_1(x, y)$ and the *query* pattern $s_2(x, y)$ are displayed side-by-side in the same input plane [5].

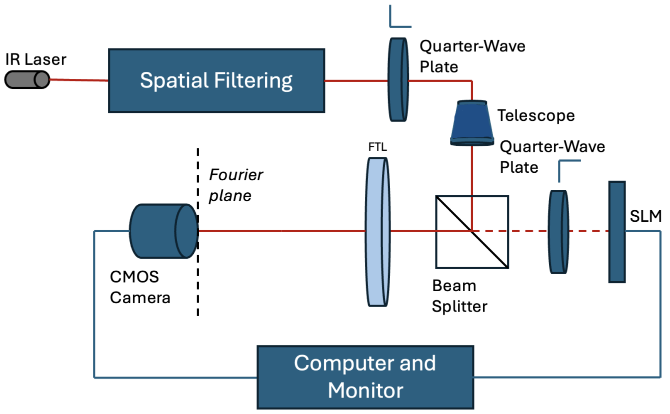


Figure 4: Optical setup for the joint transform correlator. The SLM displays the reference and query patterns side by side.

Operational principle. Let the two patterns be centred on the SLM display (phase modulation only) at $\pm x_0$ along the horizontal axis, i.e. $s_1(x + x_0, y)$ and $s_2(x - x_0, y)$. A $1f$-configuration lens (focal length f) produces at its back focal plane the joint Fourier transform, given by:

$$G(\alpha, \beta) = S_1\left(\tfrac{\alpha}{\lambda f}, \tfrac{\beta}{\lambda f}\right) e^{-i2\pi x_0 \alpha/\lambda f} + S_2\left(\tfrac{\alpha}{\lambda f}, \tfrac{\beta}{\lambda f}\right) e^{+i2\pi x_0 \alpha/\lambda f},$$

where (α, β) are spatial-frequency coordinates and S_j denotes the Fourier transform of s_j. A CMOS camera records and stores the *joint power spectrum* $|G|^2$; its intensity contains four terms:

$$|G|^2 = |S_1|^2 + |S_2|^2 + S_1 S_2^* e^{-i4\pi x_0 \alpha/\lambda f} + S_2 S_1^* e^{+i4\pi x_0 \alpha/\lambda f}.$$

The SLM displays the joint power spectrum and the same lens takes the inverse Fourier transform of this intensity pattern. This produces at the fourier plane the correlation signals of the two images, given by:

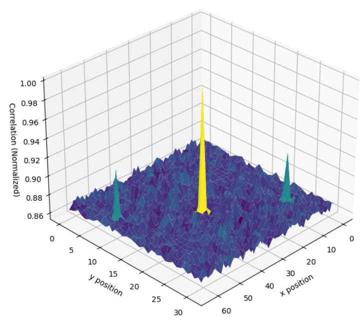
$$g(x', y') = R_{11}(x', y') + R_{22}(x', y') + R_{12}(x' - 2x_0, y') + R_{21}(x' + 2x_0, y')$$

where (x', y') are the coordinates at the correlation plane and the correlation signals are defined as:

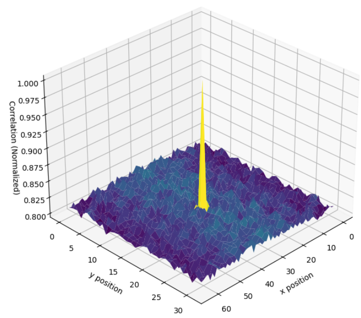
$$R_{ij}(x', y') = \int\int s_i(x' - x, y' - y) s_j(x', y')\, dx\, dy$$

The constant terms yield *autocorrelation* peaks near the optic axis, whereas the two cross-terms produce a pair of off-axis peaks located at $(\pm 2x_0, 0)$ whose heights are proportional to the cross-correlation $s_1 \star s_2$. As illustrated in Figure 5a, when the input is an image and its shifted version, the patterns are highly similar, resulting in strong and well-defined cross-correlation peaks (R_{12} and R_{21}). In contrast, Figure 5b shows the correlation between an image and an unrelated random pattern, yielding weak or undetectable peaks due to the lack of similarity.

Therefore, we just need to measure the intensity of the cross-correlation peaks to determine the similarity between the two patterns. To obtain a normalized value between 0 and 1, we can divide the intensity of the cross-correlation peaks by the intensity of the auto-correlation peak. This normalization ensures that the resulting value is independent of the absolute intensity levels and only reflects the relative similarity between the two patterns. This gives us a value of 1 when the two images are identical and 0 when they are completely different.



(a) Original image vs. shifted copy



(b) Original image vs. random image

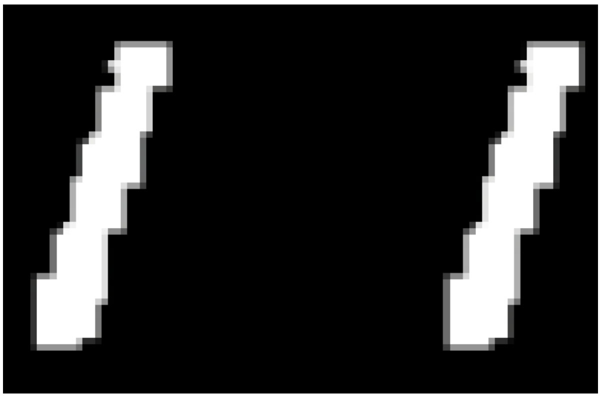
Figure 5: Normalized correlation-plane intensity produced by the joint transform correlator (with a simple Python FFT algorithm): (a) shows two strong off-axis peaks because the query is a shifted version of the reference, whereas (b) shows no significant peaks for an unrelated image.

This is the classical version of the JTC. There have been proposals of introducing some non-linearity both at the fourier plane and input plane, and that improves the performance of the JTC ([7], [5], [3], [1]).

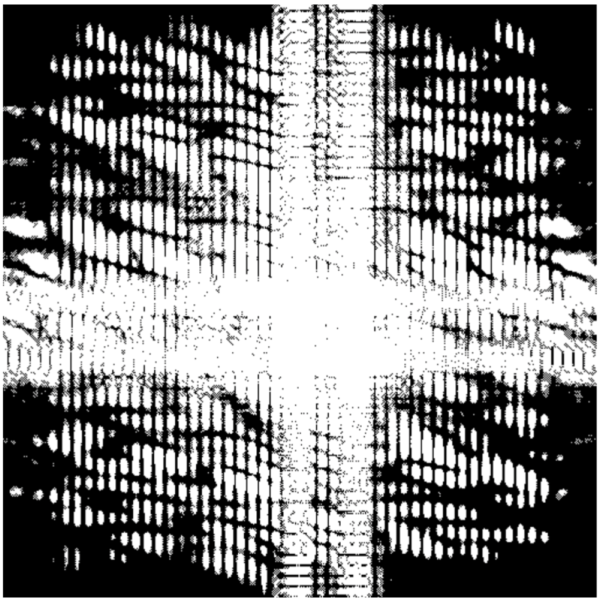
ii. Binary Joint Transform Correlator

Javidi & Horner's single–SLM Binary JTC [5] applies a hard threshold at two points of the optical cycle and works entirely with a bipolar alphabet $\{-1, +1\}$. At the input plane the grey-scale reference and query patterns are turned into a $-1/+1$ map by comparing each pixel with the median of its histogram (Fig. 6a). On a phase-only SLM those two levels translate into a 0 or π phase delay, so no optical power is wasted. After a first Fourier lens the camera records the joint power spectrum. That spectrum is thresholded in exactly the same fashion and written back to the same SLM during a second exposure (Fig. 6b), where a second Fourier lens produces the correlation plane (Fig. 6c).

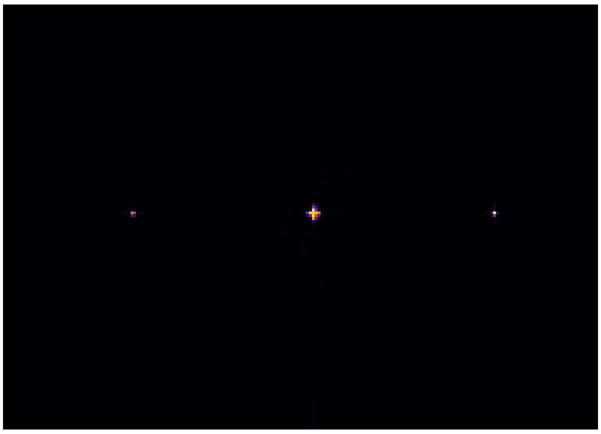
The double threshold approach has two immediate benefits. First, it transforms autocorrelation terms into sharp, delta-like spikes (Fig. 6c), which clears space on the detector for larger reference windows and removes the placement constraints typically imposed by the classical JTC. Second, it ensures the correct phase of the first harmonic is preserved through the non-linearity, while higher harmonics become both weaker and spatially separated; a simple spatial filter can then isolate the desired term. Besides those, the binary JTC has more benefits: it substantially increases the correlation-peak intensity, enhances the peak-to-sidelobe ratio, sharpens the correlation peak, and improves the signal-to-noise ratio compared to the classical JTC under the same illumination conditions.



(a) Binary input images of the JTC, notice the grid introduced by the fill-factor of the SLM.



(b) Binary joint power spectrum produced after tresholding the joint power spectrum measured by the camera.



(c) Correlation plane intensity, showing the cross-correlation peaks at the off-axis positions.

Figure 6: Python-based optical simulation of a binary-input Joint Transform Correlator, implemented with LightPipes for realistic wave-propagation and Fourier-transform modeling.

iii. JTC-based dissimilarity

Let $\mathrm{sim}(s_1, s_2) \in [0, 1]$ denote the normalised height of the cross-correlation peak produced by the joint transform correlator when the input patterns are s_1 and s_2. We quantify dissimilarity through its reciprocal,

$$d_{\mathrm{JTC}}(s_1, s_2) = \frac{1}{\mathrm{sim}(s_1, s_2)}.$$

Because $\mathrm{sim} = 1$ for identical patterns and decreases towards 0 as the patterns diverge, d_{JTC} is non-negative and symmetric (check this! they are not symmetric), but it is *not* a metric in the strict mathematical sense: it does not satisfy the triangle inequality and it is not zero for equal images. Nonetheless it serves the same practical purpose as a distance — patterns that are more alike yield a smaller d_{JTC}, and that is the most important property for the classification task.

The high R^2 value in Fig. 7 confirms that d_{JTC} preserves the rank ordering supplied by the Euclidean metric while being considerably cheaper to obtain optically, making it a suitable drop-in replacement for similarity-based classification.

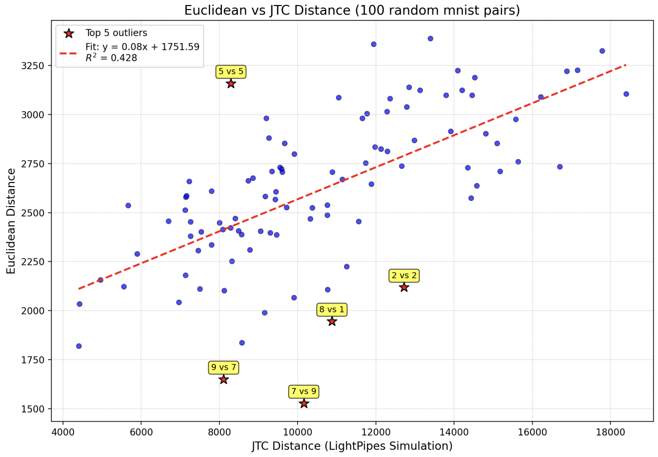


Figure 7: Scatter plot of the JTC dissimilarity versus the Euclidean distance for 100 random pairs of MNIST digits. The coefficient of determination is $R^2 = 0.428$. While the R^2 value is relatively low, multiple outliers contribute to this reduction. However, the general trend of the data is well captured by the fitted line, indicating a meaningful correlation between JTC dissimilarity and Euclidean distance.

IV. Optical implementations of the classifiers

i. Optical implementation of the CNM-C and RBF-C

We can leverage the JTC to calculate the distances instead of using the Euclidean distance. (Develop this more).

ii. Optical implementation of the QNM–C

In Sec. ii we defined the QNMC-C. There, the decision rule was formulated in terms of the *trace distance*, yet this quantity is not directly accessible in a classical–optical setup: a JTC yields a scalar similarity signal rather than the operator norm $\frac{1}{2}\|\rho - \sigma\|_1$. So in order to implement the QNM-C we need to find a method to use the JTC to obtain an estimate of the trace distance, which we can then use to classify the test samples. We will start by defining the purity of a quantum state, then the fidelity between two quantum states, and finally relate the fidelity to the trace distance. This will allow us to use the JTC to estimate the trace distance and implement the QNM-C classifier.

Purity. For a single state the purity

$$P(\rho) = \mathrm{Tr}(\rho^2)$$

ranges from 1 (pure state) to $1/d$ (maximally mixed state in Hilbert space of dimension d).

Fidelity. For two states ρ, σ the fidelity is defined as[1]

$$F(\rho,\sigma) = \left[\mathrm{Tr}\left(\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\right)\right]^2.$$

[1] See, e.g. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniv. ed., 2010.

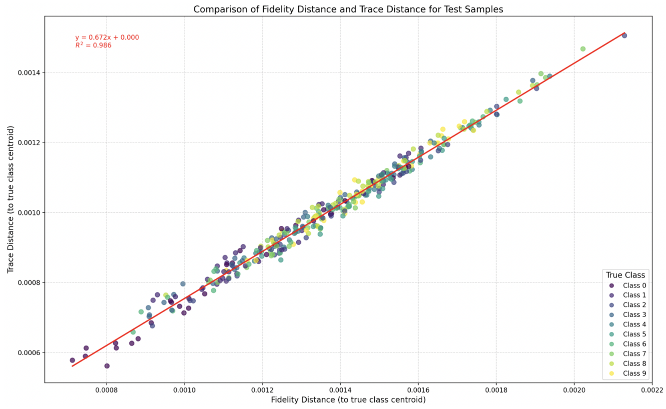


Figure 8: Trace distance vs. fidelity distance computed with the JTC for the quantum-inspired nearest-mean classifier (QNM-C) using the inverse stereographic encoding. The R^2 value of 0.986 indicates a strong linear relationship between the two measures, confirming that the JTC can effectively estimate the trace distance from the fidelity.

When both states are pure, $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\phi\rangle\langle\phi|$, this simplifies to

$$F(\rho,\sigma) = |\langle\psi \mid \phi\rangle|^2.$$

Because the JTC similarity is proportional to $\langle\psi \mid \phi\rangle$, its squared magnitude is directly proportional to the fidelity of two pure optical encodings. This is only true for pure states, for mixed states the fidelity cannot be calculated directly from the JTC similarity.

Trace distance from fidelity. The Fuchs–van de Graaf inequalities relate fidelity and trace distance D_{tr}:

$$1 - \sqrt{F(\rho,\sigma)} \;\le\; D_{\mathrm{tr}}(\rho,\sigma) \;\le\; \sqrt{1 - F(\rho,\sigma)}.$$

If one of the states is pure, the trace distance can be expressed in terms of the fidelity:

$$D_{\mathrm{tr}}(\rho,\sigma) = \sqrt{1 - F(\rho,\sigma)}.$$

Denoting the JTC similarity output by $\mathrm{sim}(\rho,\sigma) = |\langle\psi \mid \phi\rangle|$, the trace distance between two quantum centroids ϱ_k and ϱ_j is therefore

$$D_{\mathrm{tr}}(\varrho_k,\varrho_j) = \sqrt{1 - \mathrm{sim}(\varrho_k,\varrho_j)^2}.$$

It can be seen in Figure 8 that, for the inverse stereographic encoding, the trace distance and fidelity are strongly correlated, with an R^2 value of 0.986. This confirms that the JTC can effectively estimate the trace distance from the fidelity, allowing us to use the JTC to implement the QNM-C classifier. This calculation of the trace distance using the JTC is not exclusive to the QNM-C classifier, it can be used by any quantum-inspired classifier that uses the trace distance to perform classification.

Uhlmann's theorem. The main difficulty in using this approach to compute the trace distance is that we can only evaluate the trace distance directly for pure states, whereas quantum centroids are generally mixed states. However, Uhlmann's theorem allows us to calculate the fidelity between two mixed states by using their purifications. A purification is the process of representing a mixed state as a pure quantum state of higher-dimensional Hilbert space. Uhlmann's theorem states that for any two density operators ρ and σ, there exist purifications $|\psi_p\rangle$ of ρ and $|\phi_p\rangle$ of σ such that

$$F(\rho,\sigma) = \max_{|\psi_p\rangle,|\phi_p\rangle} |\langle\psi_p|\phi_p\rangle|^2$$

where the maximum is taken over all purifications of ρ and σ. This means that we can compute the trace distance between two mixed states by computing the fidelity between their purifications, which can be done using the JTC.

Therefore, even though the quantum centroids are generally mixed states, we can still compute the trace distance between the quantum centroid and input vector by finding a suitable purification of the quantum centroid (the input vector is already pure). This allows us to use the JTC to compute the trace distance between the quantum centroid and the input vector, which can then be used to classify the input vector.

Choosing the correct encoding. The encoding used has a significant impact on the performance of the QNM-C classifier, as can be seen in table 2. For the optical version this is especially true, since the choice of the encoding impacts the purity of the quantum centroids (table **??**), which in turn affects the fidelity and trace distance calculations.

Table 1: Centroid purity for each MNIST class under the three encodings described in sec..

Class	Standard	Informative	Stereographic
0	0.1776	0.4623	1.0000
1	0.3186	0.5714	1.0000
2	0.0901	0.3682	1.0000
3	0.1052	0.3842	1.0000
4	0.0990	0.3863	1.0000
5	0.0830	0.3270	1.0000
6	0.1321	0.4133	1.0000
7	0.1493	0.4320	1.0000
8	0.0791	0.3454	1.0000
9	0.1065	0.3758	1.0000

iii. Optical implementation of the RBF-C and CNM-C classifiers

To implement the RBF-C and CNM-C classifier optically, we can just swap the Euclidean distance for the JTC dissimilarity. This means that we can use the JTC to compute the dissimilarity between the input vector and the centroids of each class, and then assign the class with the lowest dissimilarity. This is a straightforward implementation, as the JTC can be used to compute the dissimilarity in a parallel and efficient manner.

iv. Computational simulations

We ran all the classifiers using the MNIST dataset using PCA to reduce the dimensionality to 50 features. For the RBF-C we used a reduced version of the dataset (6000 training samples and 1000 test samples) to speed up the training process, as the RBF-C is computationally expensive. The CNM-C and QNM-C were trained on the full dataset (60000 training samples and 10000 test samples). The JTC was simulated using

Table 2: Balanced accuracy (% mean ± std.) on MNIST. For the classical classifiers (RBF-C and CNM-C) the encoding is not applicable, so we leave it empty. For the quantum-inspired classifiers (QNM-C) we show the results for the three encodings described in Sec. ii. The Fidelity distance is being calculated using the JTC, as described in Sec. ii. The best results for each encoding are in **bold**.

Classifier	Metric	Encoding	Balanced Acc.
RBF-C	Euclidean	–	–
RBF-C	JTC	–	–
CNM-C	Euclidean	–	80.38 ± 0.38
CNM-C	JTC	–	72.42 ± 0.55
QNM-C	Trace	Standard	$\mathbf{85.84 \pm 0.38}$
QNM-C	Fidelity	Standard	78.77 ± 0.34
QNM-C	Trace	Stereographic	80.41 ± 0.41
QNM-C	Fidelity	Stereographic	80.74 ± 0.44
QNM-C	Trace	Informative	81.26 ± 0.37
QNM-C	Fidelity	Informative	$\mathbf{82.39 \pm 0.40}$

V. Experimental results

The optical setup used for the experiments consisted of a reflective, phase-only spatial light modulator (SLM) with a pixel pitch of 8 μm, illuminated by a continuous-wave, diode-pumped solid-state laser at a wavelength of $\lambda = 532$ nm. The laser beam was expanded and collimated to achieve a uniform illumination across the SLM. A Fourier-transform lens with a focal length of $f = 200$ mm was used to perform the optical Fourier transform. The CMOS camera at the output plane had a resolution of 2048×2048 pixels and a pixel size of 5.5 μm, capturing the joint power spectrum and the correlation plane.

The input patterns were encoded onto the SLM using binary phase modulation (0 or π) to maximize diffraction efficiency while maintaining a clear mapping of input data. The fill-factor of the SLM introduced a visible grid pattern in the recorded images, resulting in diffraction effects that produced multiple correlation peaks in the Fourier plane.

Figure 9 illustrates the optical setup, including the laser, collimation optics, SLM, Fourier-transform lens, and CMOS camera.

figures/experimental_setup_placeholder.png

Figure 9: Schematic of the experimental setup for the JTC-based optical correlator. The laser beam is expanded and collimated, modulated by the SLM, and Fourier-transformed onto the CMOS camera.

The correlation plane exhibited distinct off-axis peaks corresponding to the cross-correlation between the reference and query patterns. When the query pattern matched the reference (e.g., a shifted version), these peaks were sharp and intense, demonstrating strong similarity. In contrast, when the query was unrelated to the reference, the correlation peaks were weak or absent, confirming low similarity. The presence of the SLM's fill-factor grid caused diffraction effects, producing additional spots in the correlation plane, as expected from the pixelated structure of the modulator.

These observations confirm the effectiveness of the optical JTC implementation in detecting similarity between input patterns and validate its use in the proposed classifiers.

VI. Conclusion

Weird things: for the quantum nearest mean classifier applying the standard scaller yields worse results, I don't know why

i. Future Work

- Paralell JTC - Other optical architectures - Explore other quantum inspired classifiers that can leverage optical implementations

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